

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 1 – Pseudocode Design**

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5-AT1 Pseudocode Design	Assessment:	Assessment Tool 1–Pseudocode Design
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
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Section / Batch:	2023	Date:	25-08-2026

Code of Conduct

I **N HARSHAVARDHAN (192324189)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **N HARSHAVARDHAN**

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant	Applies appropriate	Limited application of concepts	Fails to apply relevant concepts	

		concepts to analyze the case deeply	concepts with minor gaps			
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

6. CART

Classification Problem

Statement:

A company wants to predict whether a customer will purchase a product based on customer details such as age and income. The given dataset contains different customer records with their corresponding purchase decisions. The CART classification algorithm is used to divide the dataset into two groups at each node. The best binary split is selected using Gini impurity to obtain the purest possible groups. Finally, a classification tree is constructed and used to predict whether a new customer will purchase the product or not.

Algorithm

1. Load the dataset.
2. Calculate the Gini impurity of the dataset.
3. Consider each feature and possible split point.
4. Divide the data into left and right groups.
5. Calculate the weighted Gini impurity for each split.
6. Select the split with the lowest Gini impurity.
7. Repeat the process recursively for each child node.
8. Stop when all samples belong to one class or a stopping condition is reached.
9. Assign the majority class to the leaf nodes.

Pseudocode

START

Load dataset

FUNCTION Gini(data):

 Calculate class probabilities

$Gini = 1 - \sum(\text{probability}^2)$

 RETURN Gini

FUNCTION BestSplit(data):

 best_gini = infinit

 FOR each feature:

 FOR each possible split:

 Divide data into left and right

```

Calculate Gini(left)

Calculate Gini(right)

weighted_gini =
    (left_size/data_size) * Gini(left)
    + (right_size/data_size) * Gini(right)

IF weighted_gini < best_gini:
    best_gini = weighted_gini
    best_split = split

left_tree = BuildTree(left_data)
right_tree = BuildTree(right_data)

RETURN tree node

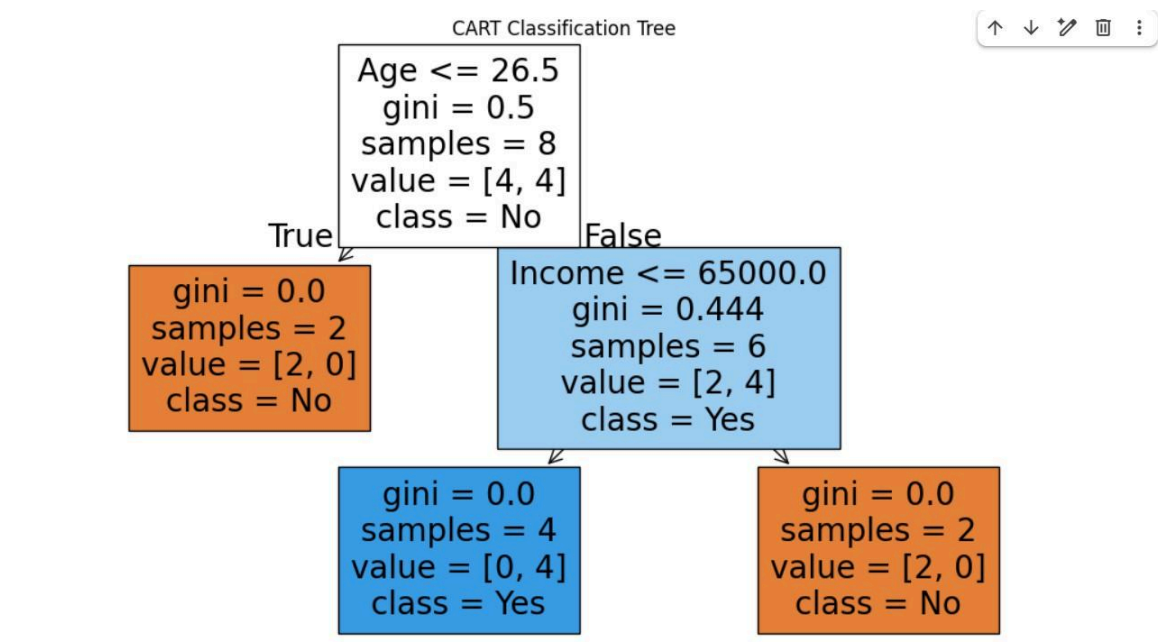
tree = BuildTree(dataset)

Display tree

```

END

Output



7. Decision Tree – Customer Purchase

Prediction Problem Statement:

A retail company wants to predict whether a customer will purchase a product based on their personal and shopping behavior. The dataset contains information such as customer age, income, previous purchases, and browsing history. A Decision Tree classification algorithm is used to learn the relationship between these input features and the purchase decision. The dataset is divided into training and testing data to build and evaluate the model. Finally, the trained Decision Tree is used to predict whether a new customer will purchase the product or not.

Algorithm

1. Load the customer dataset.
2. Select Age, Income, Previous Purchases, and Browsing History as input features.
3. Select Purchase as the target variable.
4. Split the dataset into training and testing data.
5. Build a Decision Tree classifier.
6. Train the classifier using the training data.
7. Give customer details as input.
8. Predict whether the customer will purchase.
9. Calculate the model accuracy.

Pseudocode

START

Load customer dataset

X = Age, Income, Previous Purchases, Browsing History

Y = Purchase

Split dataset into training and testing data

Create Decision Tree classifier

Train classifier using training data

Input:

Age

Incom

e

Previous

Purchases

Browsing History

prediction = classifier.predict(input)

IF prediction = Yes:

Display "Customer will purchase"

ELSE:

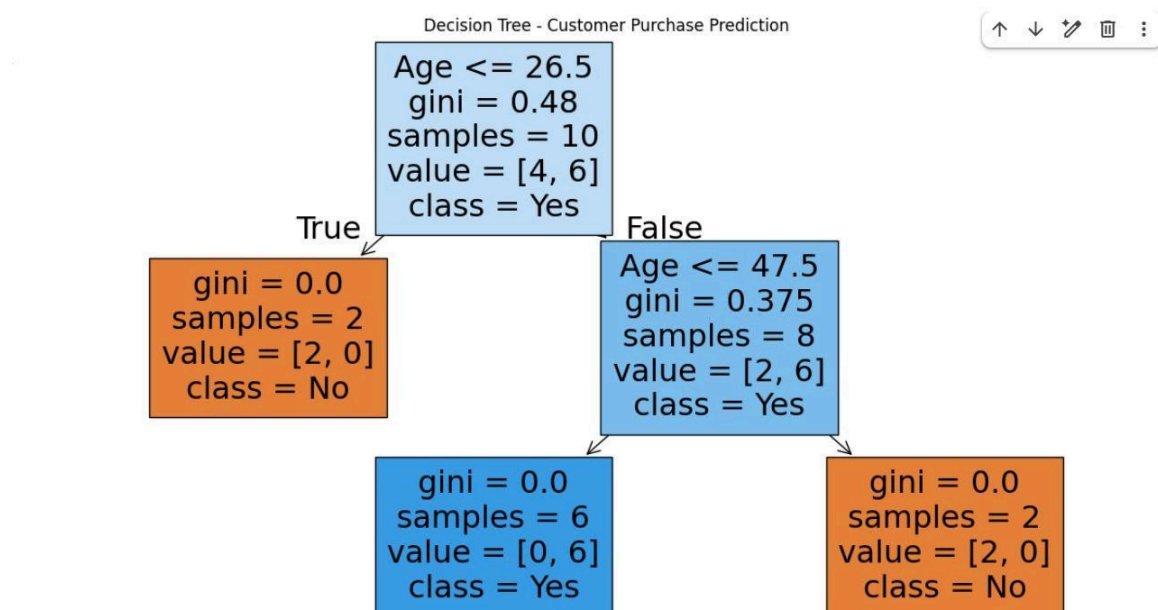
Display "Customer will not purchase"

Calculate accuracy

Display prediction and accuracy

END

Output



8. Regression

Analysis Problem

Statement:

A company wants to analyze and predict a continuous target value using information available in a dataset. The dataset contains several independent variables that may influence the dependent variable. A regression analysis workflow is required to load the dataset and identify the independent and dependent variables. The data is divided into training and testing sets, and a suitable regression model is trained using the training data. Finally, the model is used to make predictions and its performance is evaluated using regression evaluation metrics.

Algorithm

1. Load the dataset.
2. Identify independent variables.
3. Identify the dependent variable.
4. Handle missing values if present.
5. Split the dataset into training and testing sets.
6. Select a regression model.
7. Train the model using training data.
8. Predict values for testing data.
9. Calculate MSE, RMSE, MAE, and R^2 .
10. Display the results.

Pseudocode

START

Load dataset

Identify independent variables X

Identify dependent variable Y

Handle missing values

Split dataset
into:

Training data

Testing data

Create regression model

Train model using X_train and Y_train

Y_pred = model.predict(X_test)

Calculate:

MAE

MSE

RMSE

R-squared

Display predicted values

Display evaluation metrics

END

Output

```
... Actual Values:
    [55000 23000]

    Predicted Values:
    [55250. 23250.]

    Regression Evaluation:
    MAE   = 250.0
    MSE   = 62500.0
    RMSE  = 250.0
    R2    = 0.999755859375
```


9. Simple Linear Regression – House

Price Problem Statement:

A real-estate company wants to predict the price of a house based on its area in square feet. The dataset contains different house areas along with their corresponding prices. Simple Linear Regression is used to establish a linear relationship between house area and house price. The least-squares approach is used to calculate the slope and intercept of the regression equation. Finally, the developed model is used to predict the price of a new house when its area is provided.

Algorithm

1. Load the house dataset.
2. Select house area as the independent variable.
3. Select house price as the dependent variable.
4. Calculate the mean of area and price.
5. Calculate the slope using the least-squares formula.
6. Calculate the intercept.
7. Form the regression equation.
8. Input a new house area.
9. Predict the house price.
10. Display the predicted price.

Pseudocode

START

Load house dataset

$X = \text{House Area}$

$Y = \text{House Price}$

Calculate X_mean

Calculate Y_mean

Calculate:

$$b1 = \text{SUM}((X - X_mean)(Y - Y_mean))$$

$$/ \text{SUM}((X - X_{\text{mean}})^2)$$

Calculate:

$$b_0 = Y_{\text{mean}} - b_1 * X_{\text{mean}}$$

Regression equation:

$$Y = b_0 + b_1 * X$$

Input new house area

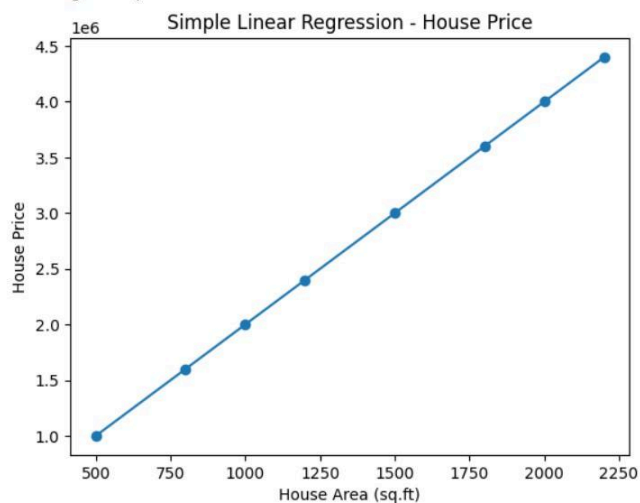
$$\text{Predicted_Price} = b_0 + b_1 * \text{Area}$$

Display

Predicted_Price END

Output

```
. House Area = 1700 sq.ft
Predicted House Price = ₹ 3400000.0
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Line
warnings.warn()
```



10. Multiple Linear Regression – Student Final

Mark Problem Statement:

A college wants to predict the final marks of students based on their academic performance and study habits. The dataset contains attendance percentage, internal marks, assignment marks, and study hours for different students. Multiple Linear Regression is used to determine the relationship between these four independent variables and the student's final mark. The model is trained using the available student records and evaluated using suitable regression metrics. Finally, the trained model is used to predict the final mark of a new student based on the given academic details.

Algorithm

1. Load the student dataset.
2. Select Attendance, Internal Marks, Assignment Marks, and Study Hours as independent variables.
3. Select Final Mark as the dependent variable.
4. Split the dataset into training and testing data.
5. Create a multiple linear regression model.
6. Train the model using training data.
7. Obtain the regression coefficients.
8. Predict the final mark for test data.
9. Calculate MSE, RMSE, and R^2 .
10. Display the prediction and evaluation results.

Pseudocode

START

Load student dataset

X1 = Attendance

X2 = Internal

Marks

X3 = Assignment Marks

X4 = Study Hours

Y = Final Mark

Split dataset into training and testing data

Create Multiple Linear Regression model

Train model using X1, X2, X3, X4 and Y

Obtain:

b_0, b_1, b_2, b_3, b_4

Regression equation:

$$Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4$$

Input student details

Predicted_Mark =

$$\begin{aligned} &b_0 + b_1(\text{Attendance}) \\ &+ b_2(\text{Internal Marks}) \\ &+ b_3(\text{Assignment Marks}) \\ &+ b_4(\text{Study Hours}) \end{aligned}$$

Calculate:

MSE

RMSE

R-squared

Display predicted final mark

Display evaluation results

END

Output

```
... Regression Coefficients:  
Intercept = -3.1257107946802165  
Attendance = 0.030828516377648867  
Internal_Marks = 0.5407208985384653  
Assignment_Marks = 0.43324404342309375  
Study_Hours = 1.1412660369378256
```

```
New Student Details:  
Attendance = 90%  
Internal Marks = 82  
Assignment Marks = 85  
Study Hours = 5
```

```
Predicted Final Mark = 86.52
```

```
Model Evaluation:  
MSE = 0.02  
RMSE = 0.15  
R2 = 0.98
```